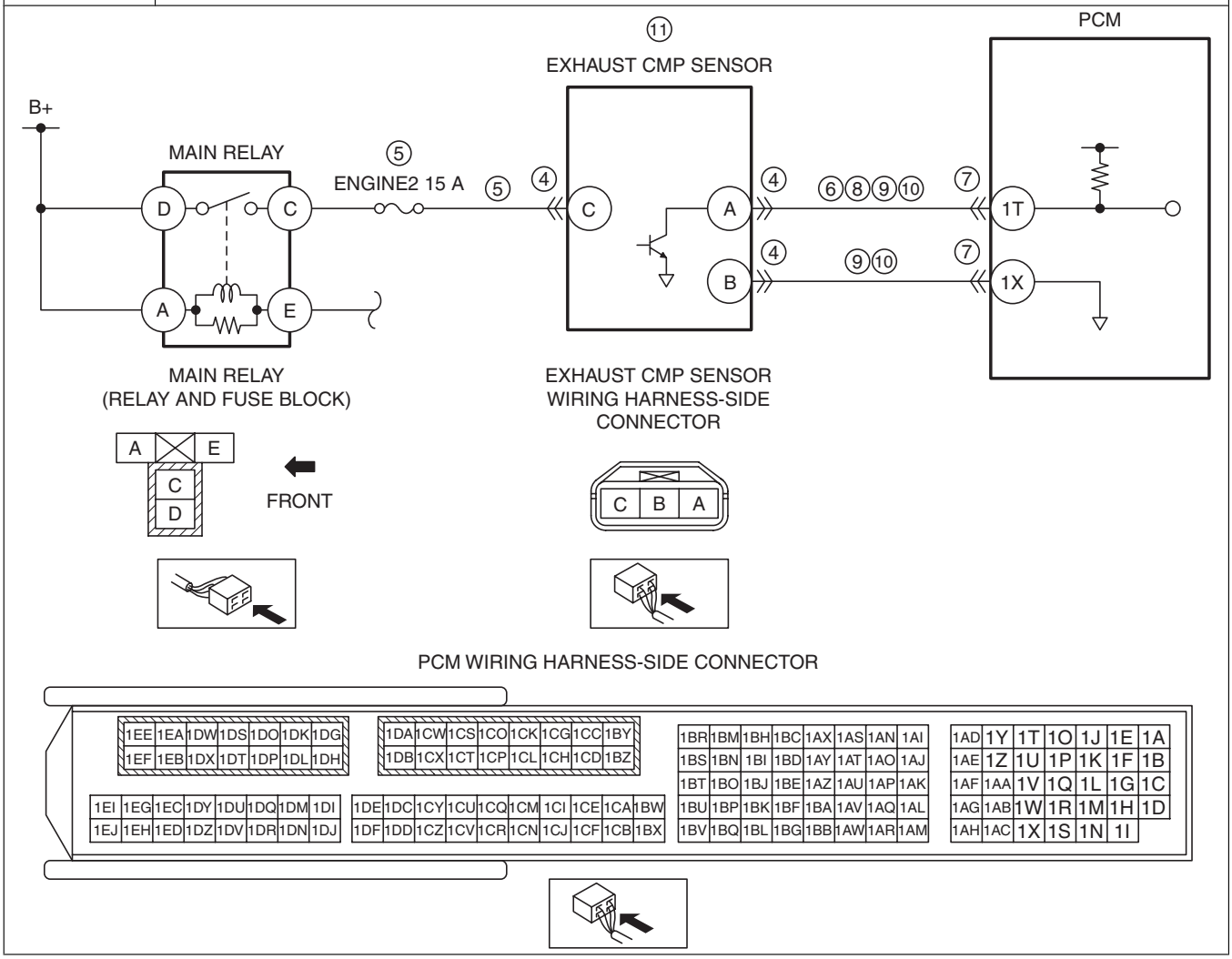
DTC P0365:00 [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

id0102q1008700

DTC P0365:00	Exhaust CMP sensor circuit problem
DETECTION CONDITION	• The exhaust CMP sensor input signal pattern, received while the crankshaft rotates **24 times**, is incorrect. • Cylinder identification is not completed while the crankshaft rotates **15 times**. **Diagnostic support note** • This is a continuous monitor (CCM). • The check engine light illuminates if the PCM detects the above malfunction condition during the first drive cycle. • FREEZE FRAME DATA/Snapshot data is available. • DTC is stored in the PCM memory.
FAIL-SAFE FUNCTION	• Stops fuel injection • Stops ignition
POSSIBLE CAUSE	• Exhaust CMP sensor connector or terminals malfunction • Short to ground or open circuit in exhaust CMP sensor power supply circuit — Short to ground in wiring harness between ENGINE2 15 A fuse and exhaust CMP sensor terminal C — ENGINE2 15 A fuse malfunction — Open circuit in wiring harness between main relay terminal C and exhaust CMP sensor terminal C • Short to ground in wiring harness between exhaust CMP sensor terminal A and PCM terminal 1T • PCM connector or terminals malfunction • Short to power supply in wiring harness between exhaust CMP sensor terminal A and PCM terminal 1T • Exhaust CMP sensor signal circuit and ground circuit are shorted to each other • Open circuit in wiring harness between the following terminals: — Exhaust CMP sensor terminal A—PCM terminal 1T — Exhaust CMP sensor terminal B—PCM terminal 1X • Exhaust CMP sensor malfunction — Exhaust CMP sensor is dirty — Exhaust CMP sensor pulse wheel malfunction • CKP sensor connector or terminals malfunction • Loose timing chain or improper valve timing • PCM malfunction

DTC P0365:00	Exhaust CMP sensor circuit problem



Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**RECORD VEHICLE STATUS AT TIME OF DTC DETECTION TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture function of the PC. • Record the FREEZE FRAME DATA/snapshot data on the repair order.	—	Go to the next step.
2	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
3	**VERIFY RELATED PENDING CODE AND/OR DTC** • Switch the ignition off, then ON (engine off). • Perform the Pending Trouble Code Access Procedure and DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Are any other PENDING CODEs and/or DTCs present?	Yes	Go to the applicable PENDING CODE or DTC inspection. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	Go to the next step.

STEP	INSPECTION		ACTION
4	**INSPECT EXHAUST CMP SENSOR CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the exhaust CMP sensor connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 14.
		No	Go to the next step.
5	**INSPECT EXHAUST CMP SENSOR POWER SUPPLY CIRCUIT FOR SHORT TO GROUND OR OPEN CIRCUIT** • Verify that the exhaust CMP sensor connector is disconnected. • Switch the ignition ON (engine off). **Note** • Another DTC may be stored by the PCM detecting an open circuit. • Measure the voltage at the exhaust CMP sensor terminal C (wiring harness-side). • Is the voltage **B+**?	Yes	Go to the next step.
		No	Inspect the ENGINE2 15 A fuse. • If the fuse is blown: — Refer to the wiring diagram and verify whether or not there is a common connector between ENGINE2 15 A fuse and exhaust CMP sensor terminal C. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. • Replace the fuse. • If the fuse is damaged: — Replace the fuse. • If the fuse is normal: — Refer to the wiring diagram and verify whether or not there is a common connector between main relay terminal C and exhaust CMP sensor terminal C. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 14.
6	**INSPECT EXHAUST CMP SENSOR SIGNAL CIRCUIT FOR SHORT TO GROUND** • Verify that the exhaust CMP sensor connector is disconnected. • Switch the ignition off. • Inspect for continuity between exhaust CMP sensor terminal A (wiring harness-side) and body ground. • Is there continuity?	Yes	If the short to ground circuit could be detected in the wiring harness: • Refer to the wiring diagram and verify whether or not there is a common connector between exhaust CMP sensor terminal A and PCM terminal 1T. **If there is a common connector:** — Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. — Repair or replace the malfunctioning part. **If there is no common connector:** — Repair or replace the wiring harness which has a short to ground. If the short to ground circuit could not be detected in the wiring harness: • Replace the PCM (short to ground in the PCM internal circuit). (See PCM REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) Go to Step 14.
		No	Go to the next step.

STEP	INSPECTION		ACTION
7	**INSPECT PCM CONNECTOR CONDITION** • Disconnect the PCM connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 14.
		No	Go to the next step.
8	**INSPECT EXHAUST CMP SENSOR SIGNAL CIRCUIT FOR SHORT TO POWER SUPPLY** • Verify that the exhaust CMP sensor and PCM connectors are disconnected. • Switch the ignition ON (engine off). **Note** • Another DTC may be stored by the PCM detecting an open circuit. • Measure the voltage at the exhaust CMP sensor terminal A (wiring harness-side). • Is the voltage **0 V**?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between exhaust CMP sensor terminal A and PCM terminal 1T. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to power supply. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to power supply. Go to Step 14.
9	**INSPECT EXHAUST CMP SENSOR SIGNAL CIRCUIT AND GROUND CIRCUIT FOR SHORT TO EACH OTHER** • Verify that the exhaust CMP sensor and PCM connectors are disconnected. • Switch the ignition off. • Inspect for continuity between exhaust CMP sensor terminals A and B (wiring harness-side). • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Exhaust CMP sensor terminal A—PCM terminal 1T • Exhaust CMP sensor terminal B—PCM terminal 1X **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to each other. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to each other. Go to Step 14.
		No	Go to the next step.
10	**INSPECT EXHAUST CMP SENSOR CIRCUIT FOR OPEN CIRCUIT** • Verify that the exhaust CMP sensor and PCM connectors are disconnected. • Inspect for continuity between the following terminals (wiring harness-side): — Exhaust CMP sensor terminal A—PCM terminal 1T — Exhaust CMP sensor terminal B—PCM terminal 1X • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Exhaust CMP sensor terminal A—PCM terminal 1T • Exhaust CMP sensor terminal B—PCM terminal 1X **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 14.
11	**INSPECT EXHAUST CMP SENSOR** • Inspect the exhaust CMP sensor. (See CAMSHAFT POSITION (CMP) SENSOR INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Is there any malfunction?	Yes	Replace the exhaust CMP sensor, then go to Step 14. (See CAMSHAFT POSITION (CMP) SENSOR REMOVAL/ INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	Go to the next step.
12	**INSPECT CKP SENSOR CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the CKP sensor connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 14.
		No	Go to the next step.

STEP	INSPECTION		ACTION
13	**VERIFY ASSEMBLY CONDITION OF TIMING CHAIN** • Verify the condition of the timing chain assembly (exhaust valve timing, looseness, jumping). (See TIMING CHAIN REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Is there any malfunction?	Yes	Repair or replace the malfunctioning part. Assemble the timing chain using the correct timing, then go to the next step. (See TIMING CHAIN REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	Go to the next step.
14	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See AFTER REPAIR PROCEDURE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Start the engine and race it. • Perform the KOER self test. (See KOEO/KOER SELF TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) Go to the next step.
		No	Go to the next step.
15	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	DTC troubleshooting completed.